

Power BI Project Requirement Document: TVS Motors Order Tracking Dashboard

Business Context:

TVS Motors is a two-wheeler manufacturer with a wide dealer network across Tamil Nadu. The business operates with a centralized order processing system, where each dealer places orders on behalf of customers for various bike models. The manufacturing process is stage-based (Quotation, Confirmation, Engine Setup, Body Setup, Final Assembly, QC, Delivery Start, Delivered), and each order can take 30-45 days to complete. The business wants to track delays, dealer performance, customer behavior, and model-wise trends using Power BI.

Objective:

To monitor dealer performance, order fulfillment stages, delay patterns, and customer insights using a centralized Power BI dashboard with advanced DAX features and dynamic KPIs.

Report-wise KPIs and Metrics

1. Dealer Performance Report

- **Total Orders Per Dealer** – Number of orders placed by each dealer.
- **Avg Delivery Time** – Average days between order and delivery.
- **On-Time Delivery %** – Percentage of orders delivered on or before expected date.
- **Delayed Orders Count** – Number of orders delivered late.
- **Top/Bottom Performing Dealers** – Based on SLA performance and delivery efficiency.
- **Dealer SLA Compliance Score** – Weighted score based on delivery timelines.

2. Order Fulfillment Tracker

- **Orders in Each Stage** – Count of orders currently in each delivery stage.
- **Completed Orders** – Orders that reached 'Delivered' stage.
- **In-Progress Orders** – Orders yet to be delivered.
- **Stage Duration (Average)** – Time taken at each stage from the 'Stages' field.
- **Recent Stage Update Timestamp** – Last updated timestamp to track freshness.
- **Fulfillment SLA Breach Count** – Orders exceeding stage deadlines.

3. Model-Wise Sales Performance

- **Total Orders by Model** – Number of orders for each bike model.
- **Revenue Estimation** – Sum of On_Road_Price × Quantity (assumed 1 per order).
- **Most Preferred Categories** – Category-wise order split.
- **Fuel Type Comparison** – Petrol vs Electric model share.

- **Avg Delivery Time by Model** – Model-wise fulfillment efficiency.

4. Customer Insights Report

- **Repeat Customers Count** – Customers with more than one order.
- **Orders by Gender** – Male vs Female buyers.
- **Customer Age Band Distribution** – Age-group-wise order count.
- **City-wise Customers** – Orders segmented by customer city.
- **Customer Loyalty Index** – Based on frequency and recency of purchases.

5. Monthly Trend & SLA Report

- **Monthly Order Volume** – Count of orders each month.
- **Monthly On-Time %** – SLA performance trend.
- **Avg Monthly Delivery Days** – Timeline benchmark analysis.
- **Cumulative Orders YTD** – Running total across year.
- **Monthly Delayed Orders** – Count of late deliveries by month.



Power BI Features to Use:

- Power Query for parsing stage data
- Date Table creation using CALENDAR()
- DAX Variables and SWITCH logic
- Dynamic Metrics using SELECTEDVALUE()
- Bookmarks & Tooltips for drill-down interactions
- Field Parameters for metric switching
- Optional Role-Level Security (Dealer-based access)



Final Output:

The Power BI dashboard will enable the business to: - Track real-time stage-wise order status - Measure delay patterns and bottlenecks - Evaluate model-wise and dealer-wise performance - Identify repeat buyers and analyze customer behavior - Improve SLA performance and streamline logistics

This dashboard will support TVS operations, sales, and supply chain teams in informed decision-making.